WASP-52b

R-0406

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-52_181213` · created 2026-07-19T05:10:07+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2458465.5901 +/- 0.0016 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.204 +/- 0.01
Transit depth (Rp/Rs)^2	4.17 +/- 0.41 [%]
Orbital Inclination (inc)	87.94 +/- 1.3
Airmass coefficient 1 (a1)	1.38 +/- 0.013
Airmass coefficient 2 (a2)	-0.234 +/- 0.024
Scatter in the residuals of the lightcurve fit is	0.92 %
Best Comparison Star	None
Optimal Aperture	4.82
Optimal Annulus	7.94
Transit Duration (day)	0.0579 +/- 0.0022

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.204 ± 0.01 vs 0.1646 (deviation 3.94σ)
Duration fit vs published	0.0579 d vs 0.0758 d (deviation 8.15σ)
Mid-time O–C	-10.7 min
Depth SNR	20.4
Residual scatter	0.92 %

Light curve

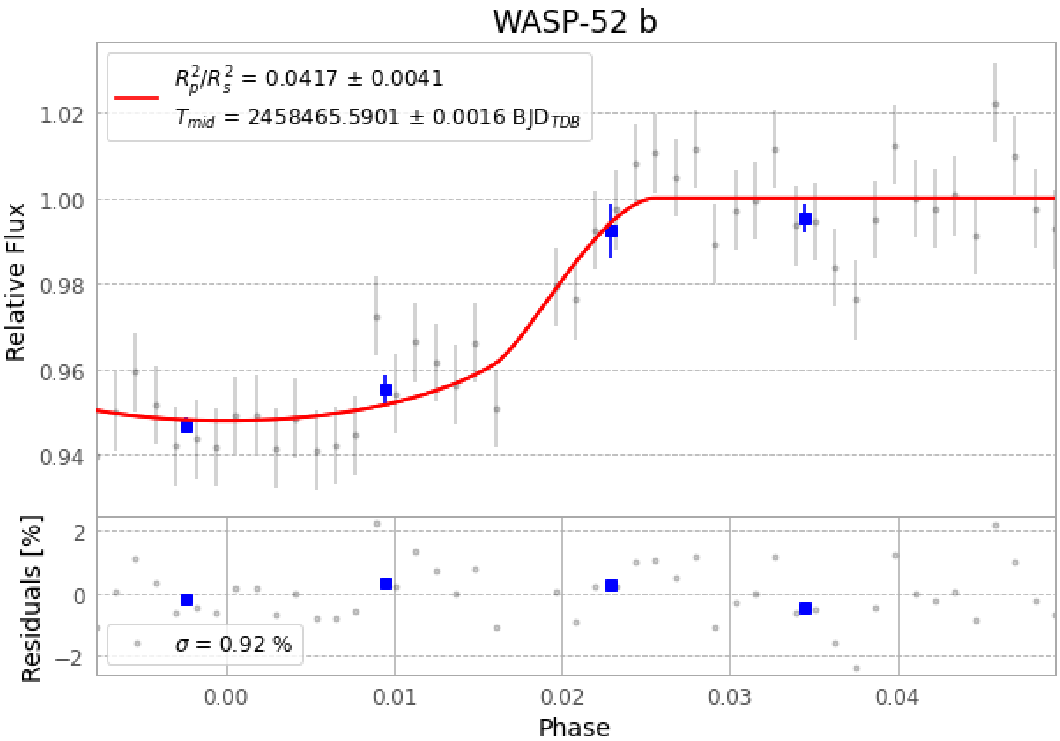

Figure 1 — FinalLightCurve_WASP-52 b_2018-12-12.png (SHA-256 35373ce12a59b752…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [272, 306], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 6c1e130a4395ca75…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 39d9b80`

Data provenance

49 FITS frames (32 MB), `WASP-52181213014812.FITS` → `WASP-52181213041212.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-52-181213.log (db4efecacd70…), FinalLightCurve_WASP-52 b_2018-12-12.png (35373ce12a59…), FinalLightCurve_WASP-52 b_2018-12-12.csv (d240783c9c41…)

Compute resources

Wall time 120.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Generated 2026-07-19 05:10Z.